CAPSTONE PROJECT UPDATE · DAE CYBERSECURITY PROGRAM

AI-Agentic SOC triage engine: build, tools and evidence

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The triage engine and its detection stack

Wazuh, Suricata, TheHive and Wireshark surround an agent that triages alerts first.

Identity

Mock, Okta-shaped API: users and login history

SIEM

Real Wazuh 4.14.7 in Docker: alerts and history

EDR

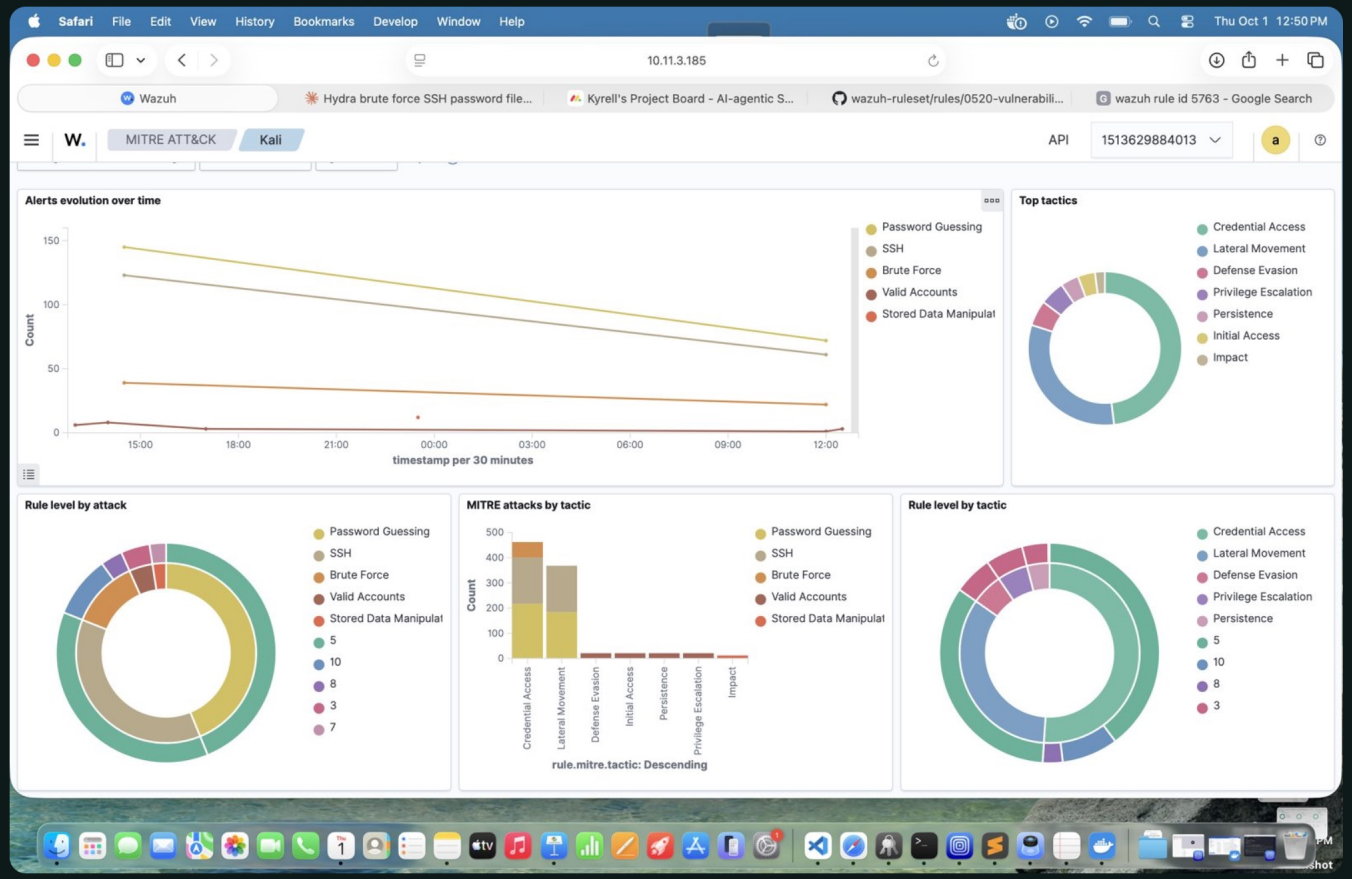
Mock, CrowdStrike-shaped API: process, network, file telemetry

LangGraph supervisor

MITRE timeline and risk summary

Human approval gate

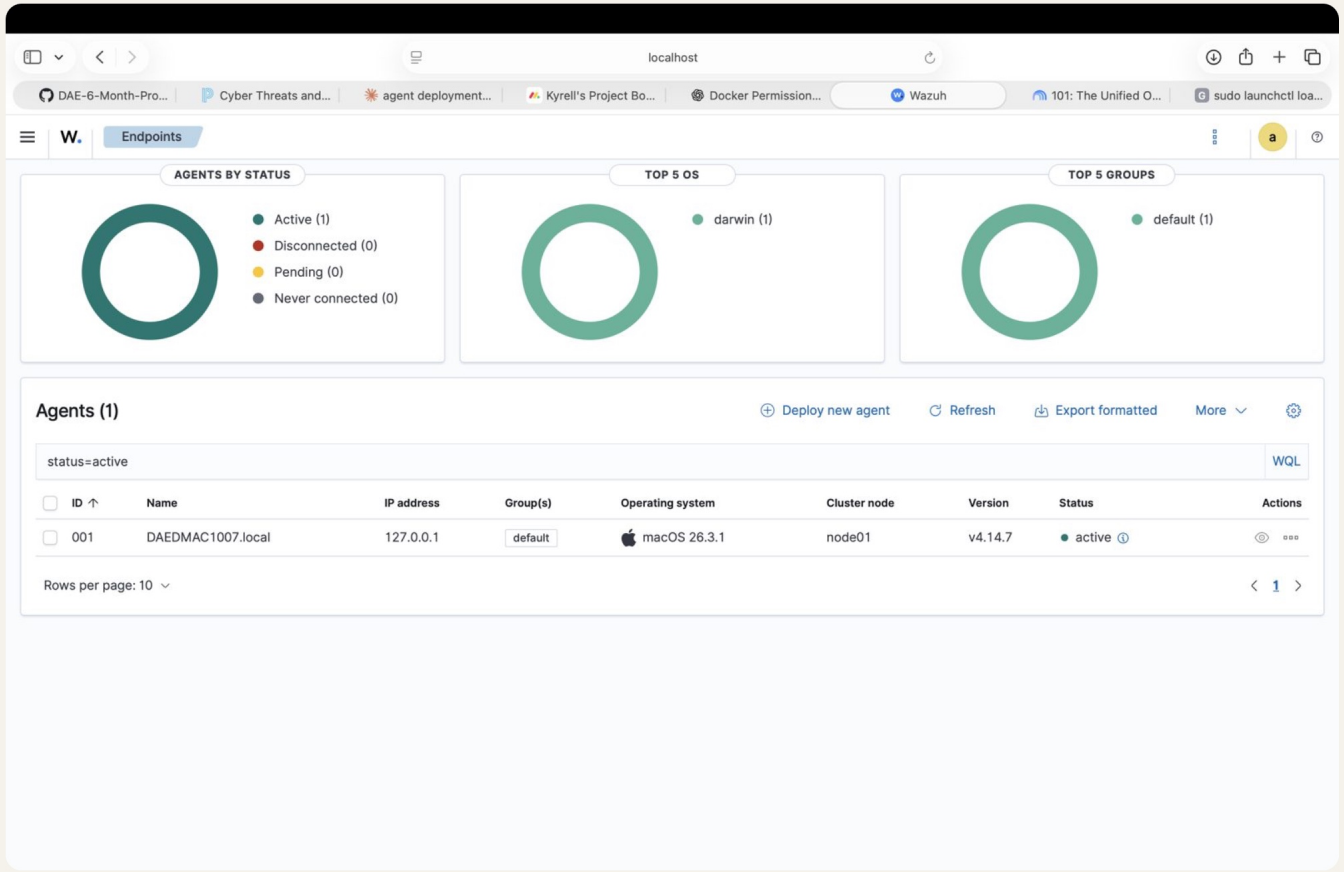
Wazuh dashboard: the analyst's single view



What the dashboard does

- Security events and rule levels in one console
- MITRE ATT&CK view: password guessing, SSH, brute force
- Top tactics: Credential Access, Lateral Movement
- Same indexer data feeds the triage agent

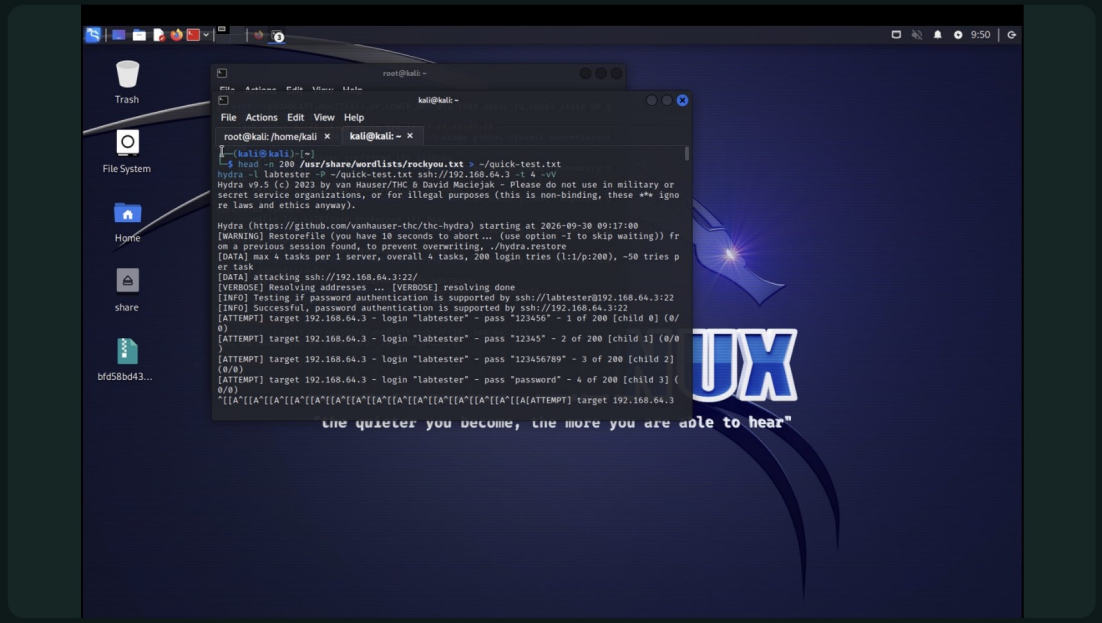
Wazuh agents: eyes on every endpoint



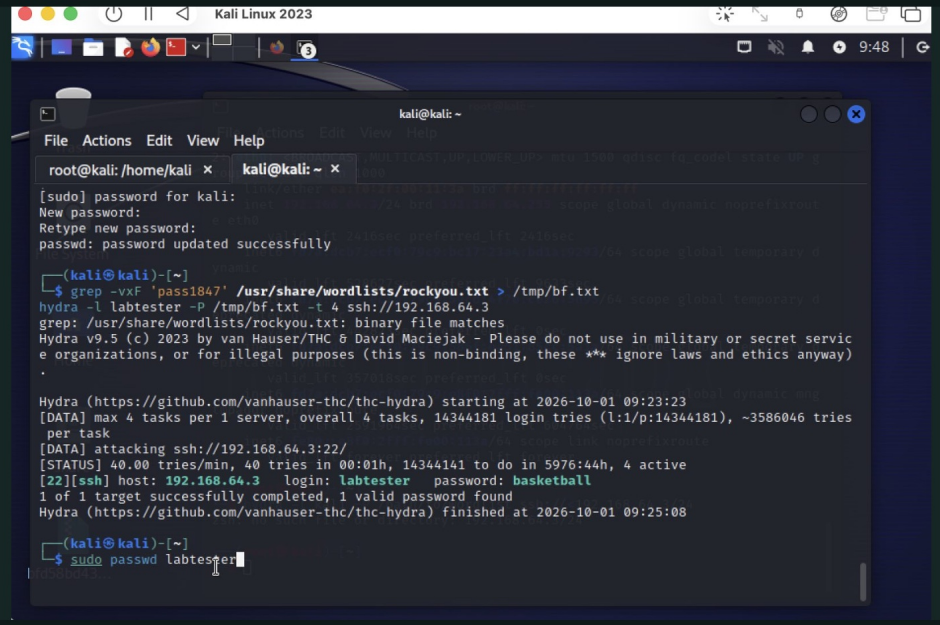
What agents do

- Collect logs, file changes and inventory
- Stream events to the manager over TCP 1514
- Enrol over TCP 1515
- Live fleet: Kali 002, Windows 007, Ubuntu 008

Kali Linux VM: attack and detection



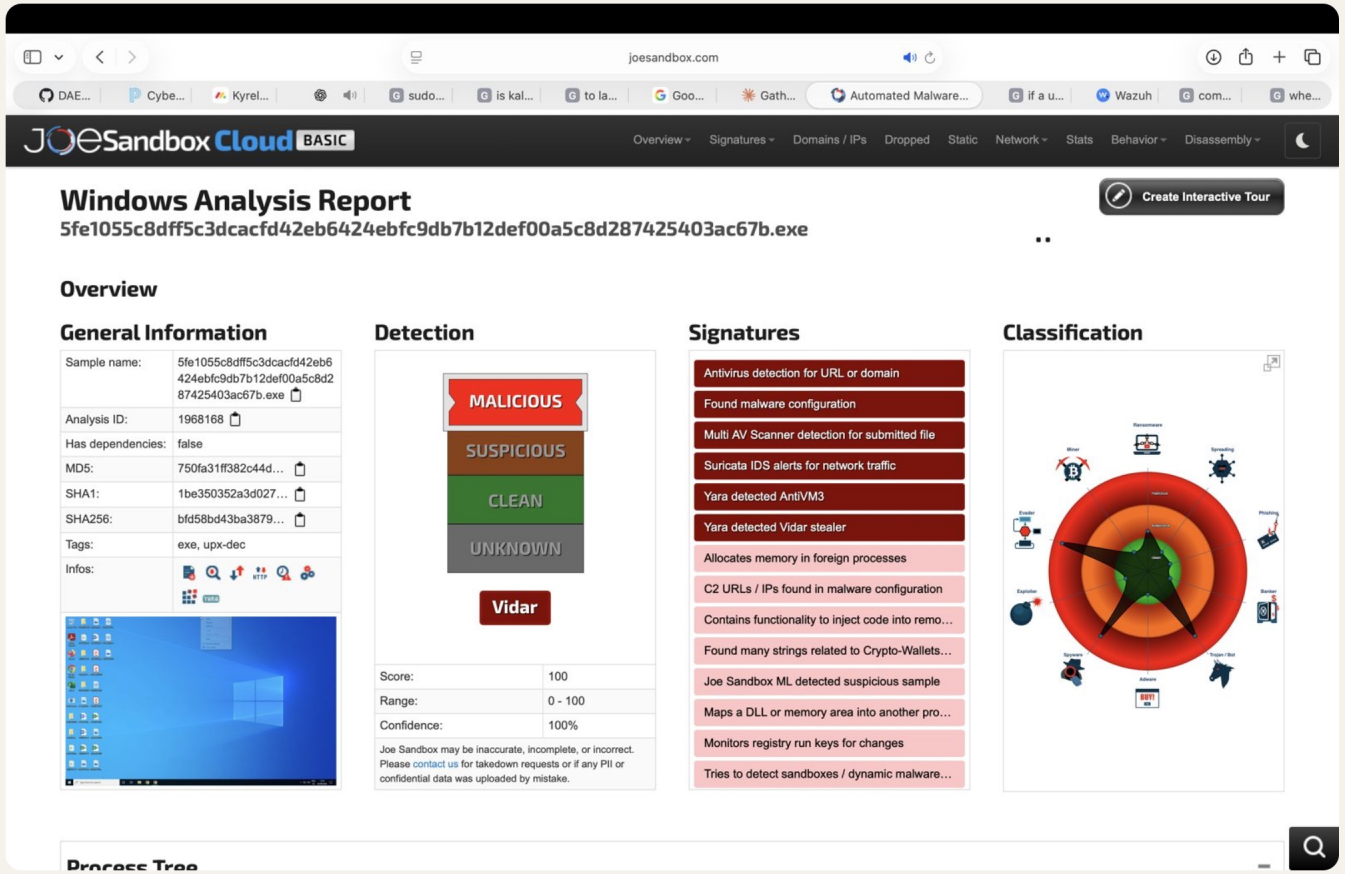
Hydra brute-force run against the Kali endpoint



Rule 40112, level 12: failures followed by a success

Kali runs in UTM with Wazuh agent 002, and acts as both target and attack source.

Endpoint detection on the Windows agent



Endpoint detection

- Rootcheck found NTFS alternate data streams
- File integrity monitoring watches file changes
- Vulnerability detection: 7 CVEs on agent 007
- Anomaly, not confirmed malware, until verified

A second line of defense around the setup

Suricata

Network IDS watches traffic. Rule ET SCAN Potential SSH Scan fired on replayed traffic.

Status: offline replay, not yet in Wazuh

TheHive

Case management. Each alert becomes a tracked case with tasks, observables and a timeline.

Status: running, connector not wired yet

Wireshark

Packet tracing. Capture filters at capture time, display filters at analysis time.

Status: verified on macOS, Kali analysis done

Wazuh sees hosts. Suricata and Wireshark see the wire. TheHive tracks the response.

One real case ties it together

SOC-CASE-2026-0142: SSH brute force that succeeded against the Kali endpoint (T1110, T1078)

4th
password attempt accepted

71 ms
session length, no activity logged

~2 min
detection to declaration

High
severity, scored with the matrix

Wazuh rule 40112 fired, the case was scored, the source blocked, and isolation was declined with a written reason. The source address was the lab host itself, a self-generated test.

How the rubric feeds the capstone

Rubric deliverable	Evidence in the repo	Role in the capstone
Security policy, incident response plan	Comprehensive Security Policy, IR Plan	Human approval gate; containment never auto-executed
Encryption and hashing	AES-256 and SHA-256 demo	Secrets stay in .env; FIM integrity checks
Vulnerability and threat intel	Nmap scan, Vidar analysis, OpenCTI plan	Enrichment and MITRE ATT&CK mapping
Topology, protocols, subnetting	Topology and Protocols reports	The 10.11.0.0/22 LAN hosting Wazuh
Firewall, IDS, IPS, monitoring	Firewall report, Monitor and Respond	Real alert source: Wazuh rule 100010, T1110
SOC operations, case management	IR Template, Methodology	Evidence bundle, timeline, analyst report

Next steps

Build

LangGraph supervisor agent with tool calls to the three evidence pillars

Close gaps

Wire Wazuh into TheHive, feed Suricata into Wazuh, and turn on alert notifications

Publish

Replace the portfolio project placeholders with the capstone and best rubric work

Goal: a SOC Analyst role, with the capstone as proof.